Lab - Ohms Law

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CPE11S4

Answer the following questions based on electricity and Ohm's Law. Show all steps when solving problems.

a. What are the four basic units of electricity? Provide the variable name and symbol, and unit name and symbol.

The four basic units of electricity are Voltage which is measured in Volts(V), Current measured in Ampere(A), Resistance measured in Ohms(Ω), and Power measured in Watt(W).

b. Write the equation for Ohm's Law.

The equation for Ohm's law is V = I × R

c. Re-arrange the Ohm's Law equation to solve the following:

I = ***V/R***

R = ***V/I***

d. Power is equal to voltage multiplied by current. Add the missing information in each of the following power equations.

P = V ***x I***

P = R ***x*** I^2

P = V^2 ***/ R***

e. The yellow wire connected to a power supply carries 12V. If the power supply provides 60W of power to the yellow wire, how much current is passing through the yellow wire?

If we're looking for the current, we're gonna divide the 60W to 12V resulting to 5A. So there's 5 Ampere passing through the yellow wire.

f. There are 3.3V passing through an orange power supply cable, and there are 0.25 ohms of resistance in the orange wire. How much power is supplied to the orange wire by the power supply?

If we're looking for the power, we're gonna turn 3.3V into $(3.3V)^2$ ***and divide it to 0.25 ohms resulting to 43.56W. Meaning there's 43.56W supplied to the orange wire by the power supply.***

g. A wire from the power supply is carrying 120W of power and 24A of current. How much power is supplied to the wire by the power supply?

In this situation, we're gonna use V = P/I turning this into 120/24 resulting with 5V. Therefore, there's 5V supplied to the wire by the power supply.

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